Horixon - Beamer Presentation Theme

A Custom Split-Architecture Template

Asumadu Prince

Kwame Nkrumah University of Science & Technology
Faculty of Physical and Computational Sciences
Department of Mathematics

National Institute of Mathematical Sciences

March 12, 2026

Table of Contents

1. Theme Architecture
2. Color & Font Palette
3. Math & HPC Macros
4. Code & Listings
5. Customisation Guide

Split Theme Architecture

This theme follows Beamer's recommended *split-theme* pattern. Each concern lives in its own `.sty` file:

Style Files

- `beamerthemehorixon.sty` — master loader
- `beamercolorthemehorixon.sty` — palette
- `beamerfontthemehorixon.sty` — typography

Layout Files

- `beamerinnerthemehorixon.sty` — title page, blocks, bullets
- `beameroutterthemehorixon.sty` — header, footer, background
- `environment-shortcuts.tex` — user macros

Color Palette

All colors are defined in `beamercolorthemehorixon.sty` and can be freely referenced in your slides:

- `horixonnavy` #222831 — primary background
- `horixondark` #393E46 — panels, header, footer
- `horixonaccent` #948979 — accent, rules, bullets
- `horixoncream` #DFD0B8 — text, light fills
- `horixonalert` — alerts, warnings

Standard Block

Uses `horixoncyan` header.

Alert Block

Uses `horixonalert` header.

Example Block

Uses `horixonlightnavy` header.

Mathematical Shortcuts

The shortcuts file provides common math macros out of the box.

Theorem (Convergence of Conjugate Gradient)

For $\mathbf{A} \in \mathbb{R}^{n\times n}$ symmetric positive definite, CG satisfies

$$\|\boldsymbol{e}_k\|_{\mathbf{A}} \leq 2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k \|\boldsymbol{e}_0\|_{\mathbf{A}},$$

where $\kappa = \lambda_{\max}/\lambda_{\min}$ is the condition number.

- `\vect{x}` $\rightarrow \boldsymbol{x}$ `\mat{A}` $\rightarrow \mathbf{A}$ `\norm{x}` $\rightarrow \|x\|$ `\BigO{n^2}` $\rightarrow \mathcal{O}(n^2)$
- `\inner{u}{v}` $\rightarrow \langle u, v\rangle$ `\abs{x}` $\rightarrow |x|$ `\mat{A}\T` $\rightarrow \mathbf{A}^\top$

Code Listings

Listing 1: MPI Hello World

```c
#include <mpi.h>
#include <stdio.h>

int main(int argc, char** argv) {
    MPI_Init(&argc, &argv);

    int world_size, world_rank;
    MPI_Comm_size(MPI_COMM_WORLD, &world_size);
    MPI_Comm_rank(MPI_COMM_WORLD, &world_rank);

    printf("Hello from rank %d of %d\n", world_rank, world_size);

    MPI_Finalize();
    return 0;
}
```

How to Customise

Change Colors

Edit `beamercolorthemehorixon.sty`. Redefine any `horixon*` color via `\definecolor`.

Change Fonts

Edit `beamerfontthemehorixon.sty`. Swap `helvet` for any sans-serif package.

Toggle Watermark

Pass `watermark=off` to `\usetheme` to disable the background overlay.

Add Your Logo

Pass `logo=assets/logos/yourlogo.png`. It appears in the footer right side.

Thank You

Asumadu Prince

March 12, 2026